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Check Your Knowledge: Inference Techniques

Graded Assignment • 10 min

Due Nov 16, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

Instructions

1. What is Bayesian inference primarily used for in machine learning? 1 / 1 point

- Maximizing the likelihood of observed data
- Estimating hidden variables from observed data
- Minimizing the error in predictions
- Solving optimization problems

Nice work

Correct! Bayesian inference is used for estimating hidden variables from observed data.

2. What does the posterior distribution in Bayesian inference represent? 1 / 1 point

- The distribution of the parameters given the observed data
- The marginal distribution of the data
- The prior distribution of the parameters
- The likelihood of the observed data

Nice work

Great job! The posterior distribution represents the distribution of the parameters given the observed data.

3. Which of the following is a key advantage of Bayesian inference over point estimation methods like ML and MAP? 1 / 1 point

- It incorporates prior knowledge and handles uncertainty.
- It provides a single best estimate of the parameters.
- It simplifies the computation of probabilities.
- It always results in faster computations.

Nice work

Correct! Bayesian inference incorporates prior knowledge and handles uncertainty.

4. Bayesian inference only provides point estimates of parameters. 1 / 1 point

- false
- true

Nice work

Correct! Bayesian inference provides a full posterior distribution of parameters, not just point estimates.

5. How is the posterior distribution (p(\theta \mid X)) calculated in Bayesian inference? 1 / 1 point

- By differentiating the likelihood function
- By using Bayes' theorem
- By maximizing the likelihood function
- By integrating the prior distribution

Nice work

Correct! The posterior distribution is calculated using Bayes' theorem.